

UCS659: BUILD AND RELEASE MANAGEMENT

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Course Objective: This course includes theory and lab. The course comprises four modules. The main objective of this course to help participants understand the process of build and release management.

Syllabus

Introduction to Build and Release Management: Introduction to build, understanding different phases of build and release management, introduction to release management, best practices for build and release management, concept of build abstraction and dependency abstraction.

Dependency Management: Introduction to dependency management, how to use source code repositories, managing transitive dependencies, dependency scope and discussion of various tools like Ant, Maven and Gradle.

Document and Reporting: Introduction to build document and reporting, different types of documentation, understanding site life cycle, advance site configurations and reports, generation of unit test reports, generation of code coverage reports, code coverage tools, code coverage pros and cons.

Release Cycle: To understand project release life cycle, different stages of release lifecycle, source code repositories, how to install and configure source code repositories and deploying build to production goals- prepare, perform, clean and rollback.

Laboratory Work: Setting up Maven environment and understanding POM hierarchy, creation of a project using Maven and its configurations.

Course Learning Objectives (CLO)

After completion of this course, the students will be able to:

1. Explain the basics of build and release management by learning build abstraction and declarative dependency management.
2. Describe dependency management and the associated concepts like repositories, dependency identification and scope, transitive dependencies, and examples for build tools.
3. Discuss the process of documentation and reporting, using site life cycle, site configuration and generation of unit testing and code coverage reports
4. Define release cycle and the phases of release, preparing, cleaning and performing goals.